

MongoDB Project about a file explorer/searcher

1. files (one doc $\approx$ 300 B)

 Copy

```
{
  _id      : ObjectId (Mongo gives this)
  uid      : string    # unique key = inode+mtime (or UUID)
  name     : string    # original name + extension
  ext      : string    # ".pdf"
  size     : long      # bytes
  path     : string    # full absolute path on disk
  folderId : ObjectId  # pointer to folders collection
  owner    : string    # "me" (multi-user prep)
  dateAdded : ISODate  # when we first saw it
  dateMod   : ISODate  # file-system mtime
  hash     : string    # SHA-256 (dedupe, future integrity check)
  mime     : string    # "application/pdf" (optional)
  meta     : sub-document # ext-specific goodies
    pdf : { pages : int, author : string, title : string }
    apk : { versionName : string, minSdk : int, iconGridFSId : ObjectId (thumb) }
    img : { width : int, height : int, colorSpace : string }
  lastOpen : ISODate  # updated by accessLog
}
```

→ Gives us: search by name, ext, size-range, mime, hash, last opened, author of PDF, etc.

2. folders

 Copy

```
{
  _id      : ObjectId
  name     : string    # "Documents"
  path     : string    # "/home/you/Documents" (unique)
  parentId : ObjectId|null # parent folder _id (materialised path)
  depth    : int       # how deep from root (speed filter)
  owner    : string
}
```

→ Lets us rebuild tree view, move whole branches, run Dijkstra on folder graph.

3. tags (user-defined labels)

 Copy

```
{
  _id : ObjectId
  name: string # "uni", "tax-2024", "holiday"
  owner: string
}
```

4. file_tags (many-to-many join)

 Copy

```
{
  fileId: ObjectId # files._id
  tagId : ObjectId # tags._id
}
```

→ Instant “show me everything tagged uni”.

5. textChunks (for full-text search inside files)

 Copy

```
{
  fileId : ObjectId
  chunkNo: int      # 0,1,2... (5000 chars each)
  text   : string   # actual text slice
}
```

Mongo text-index on `text` field → `{ $text: { $search: "scheduling algorithm" } }` returns fileIds.

6. accessLog (audit + recent-files + statistics)

 Copy

```
{
  fileId    : ObjectId
  user      : string
  action    : string    # OPEN, DOWNLOAD, DELETE, RENAME
  timestamp: ISODate
  ip        : string    # optional when we add phone
}
```

→ Recent list, most-used, heat-maps, TTL purge (90 days).

7. (optional) trash (soft-delete)

 Copy

```
{
  fileId      : ObjectId
  deletedAt   : ISODate
  autoPurgeAt: ISODate  // TTL index = today + 30 days
}
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